

[4]

6. Use Gauss - Seidel method to

$$\text{solve } 9x + 4y + z = -17$$

$$x - 2y - 6z = 14$$

$$x + 6y = 4$$

Perform only four iterations.

- 7 (a) Evaluate $\int_0^1 \frac{2x}{1+x^2} dx$ using Simpson's $\frac{1}{3}$ rule
dividing interval into 4 equal parts.

- (b) Solve the linear system by- Gauss elimination

$$\text{method } y + z = 2$$

$$2x + 3z = 5$$

$$x + y + z = 3$$

8. Solve $\frac{dy}{dx} = x + y$ with initial condition $y(0)=1$, by
Runge-Kutta method of IV order from $x=0$ to $x=0.4$
with $h=0.1$.

9. (a) Given $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y=1$ for $x=0$. Find y
approximately for $x=0.1$ by Euler -method.

- (b) Given values

X :	5	7	11	13	17
f(x):	150	392	1452	2366	5202

Evaluate $f(9)$ using Newton's divided difference
formula.

BCA -504(N)

Roll No.....

Total No. of Questions : 9]

[Total No. of Printed Page : 4]

BCA -504 (N)

Bachelor of Computer Applications

(Fifth Semester)

EXAMINATION, MARCH - 2021

(New Course)

Paper Fourth

NUMERICAL METHODS

Time : Three Hours]

[Maximum Marks : 75

Note : Attempt questions from all sections as directed.

Inst. : The candidates are required to answer only in serial
order. If there are many parts of a question, answer
them in continuation.

Section -A

(Short Answer Type Questions)

Note : Attempt all questions from this section. Each question
carries 3 marks.

- (a) Show that $(1+\Delta)(1-\nabla)=I$. Where Δ , ∇ , I
are forward difference, backward difference,
identity operators respectively.
- (b) Find the root of equation $xe^x - 1 = 0$ between 0
and 1 using bisection method. Perform only two
iterations.

P.T.O.

[2]

- (c) Discuss the convergence condition of Gauss-Seidel method.
- (d) Discuss rate of convergence of Newton-Raphson method.
- (e) Find $\Delta^2(x^2 + 1)$ (take $h=1$)
- (f) Find the interpolating polynomial for data
- | | | | |
|----|---|---|---|
| x: | 0 | 1 | 2 |
| y: | 0 | 5 | 2 |
- (g) Estimate the missing terms in the following table.
- | | | | | | | |
|----|-----|----|----|-----|-----|-----|
| x: | 16 | 18 | 20 | 22 | 24 | 26 |
| y: | 139 | 85 | — | 151 | 264 | 388 |
- (h) Evaluate $\int_0^{\pi/2} \sin x \, dx$ by Trapezoidal rule.
(take $h = \frac{\pi}{10}$)
- (i) Using Picard's method upto third approximation solve.

$$\frac{dy}{dx} = x + y^2, \quad y(0) = 0$$

Section B

(Long Answer Type questions)

Note : Attempt any 2 questions. Each question carries 12 marks.

Q2. Following table gives the marks secured by 100 students in the numerical analysis subject

BCA -504(N)

[3]

Range of marks :	30-40	40-50	50-60	60-70	70-80
No. of Students :	25	35	22	11	7

Use Newton's forward difference interpolation formula to find

- (i) Number of students who got more than 55 marks
- (ii) Number of student who got marks between 36 to 45.

Q3. Use stirling formula to find y_{28} , given that

$$y_{20} = 49225, y_{25} = 48316, y_{30} = 47236$$

$$y_{35} = 45926, y_{40} = 44306$$

Q4. (a) Evaluate $\sqrt{12}$ to five decimal places by Newtons iterative method.

(b) Solve $x^3 - 9x + 1 = 0$ for the root lying between 2 and 4 by Regula falsi method.

Q5. The table given below reveals. the velocity 'v' of a body during time 't'. Find its acceleration at $t = 1.1$ i.e.

find $\frac{dv}{dt}$ at $t = 1.1$

t :	1.0	1.1	1.2	1.3	1.4
v :	43.1	47.7	52.1	56.4	60.8

Section -C

(Long answer type question)

Note : Attempt any 2 questions. Each question carries 12 marks.

BCA -504(N)

P.T.O.